

Forever Pollution

WHAT ARE PFAS "FOREVER CHEMICALS"?

PFAS are man-made chemicals in thousands of everyday products. **They never break down** — not in water, soil, or the body. Scientists call them "Forever Chemicals".

Analogy: Imagine spilling oil that can never be cleaned — it spreads into rivers, farms, and tap water, forever.

WHERE DO THEY COME FROM?

Tap water

Factory — soil — groundwater

Non-stick pans

Teflon[®] releases PFAS when heated

Firefighting foam

Airports & military bases

Also in: waterproof jackets, fast food packaging, makeup, electronics.

HEALTH RISKS

Cancer Hormone disruption Weakened immunity Liver damage

REAL CASE — RHÔNE RIVER, LYON

Rhone focus: Pierre-Bénite industrial platform — south of Lyon

● Arkema releases **3.5 tonnes of PFAS/year** into the Rhône — every year since 2011.

● Wells **14 km away** via underground aquifer: **695 ng/L** — 7× EU legal limit.

● **67% of local tap water** exceeds EU limit. Communes: Lyon 7&8, Oullins, Pierre-Bénite, Feyzin, Givors. Water extraction at **Grigny & Ternay** contaminated.

2023: residents told not to eat local eggs, poultry, or Rhône fish.

THE DATA CHALLENGE

CNRS collected data from **104 agencies across Europe** — different formats, languages, and units.

952,727

Measurements

104

Different sources

287,431

Unique sites

600 MB

Cleaned dataset

WHY NOT JUST USE EXCEL OR SQL?

S

Looks the same, isn't: 104 countries stored hidden details differently â€” like 104 people describing the same photo in different languages.

SQL: `json_extract(details,'$.id_station')` returns NULL for 60 of 104 sources â€” silently drops real data, no error raised.

â€” Graph: each node stores **only what it actually has** â€” Italian nodes have `id_station` , French nodes have `site_reference` . No fixed schema to violate. Query only what exists per node.

M

Lots of holes: some sites have real measurements, others say "suspected" â€” mixing them gives wrong averages.

SQL: `AVG(pfas_sum)` silently ignores suspected sites with NULL â€” making contamination look 40% lower than it is.

â€” Graph: suspected sites exist as `PresumptiveSite` nodes **with no measurement edge** â€” still visible, still scored via **Textual Score T_k**. Nothing disappears.

G

9 map systems: mixing Lambert-93, WGS84, ED50â€” impossible to combine without conversion.

SQL: a `UNION` of Lambert-93 (`x=902714`) and WGS84 (`lat=45.76`) puts French factories in the Pacific Ocean â€” silently.

â€” Graph: all 9 systems reprojected to **WGS84 before loading**. Every node guaranteed to have correct `lat/lon` â€” `WHERE lat BETWEEN 45.5 AND 46.0` reliably returns only Lyon sites.

OUR APPROACH â€” IN PLAIN ENGLISH

Instead of forcing messy data into a rigid spreadsheet, we built a **map of connections** â€” like a social network for pollution sites, chemicals, and measurements.

Think of it like Google Maps for pollution

Every factory, river sample, and chemical is a dot. Lines show who produced what, where it was found, and how much. Ask: *"Which factory connects to the most dangerous rivers?"*

STEP 1 â€” DATA ENGINEERING: CLEANING THE MESS

Data engineering is turning raw, broken data into something reliable. Without it, every result is wrong.

Extract

Read 104 original Python scripts. Found hidden rules, country-specific logic and errors.

â†’

Transform

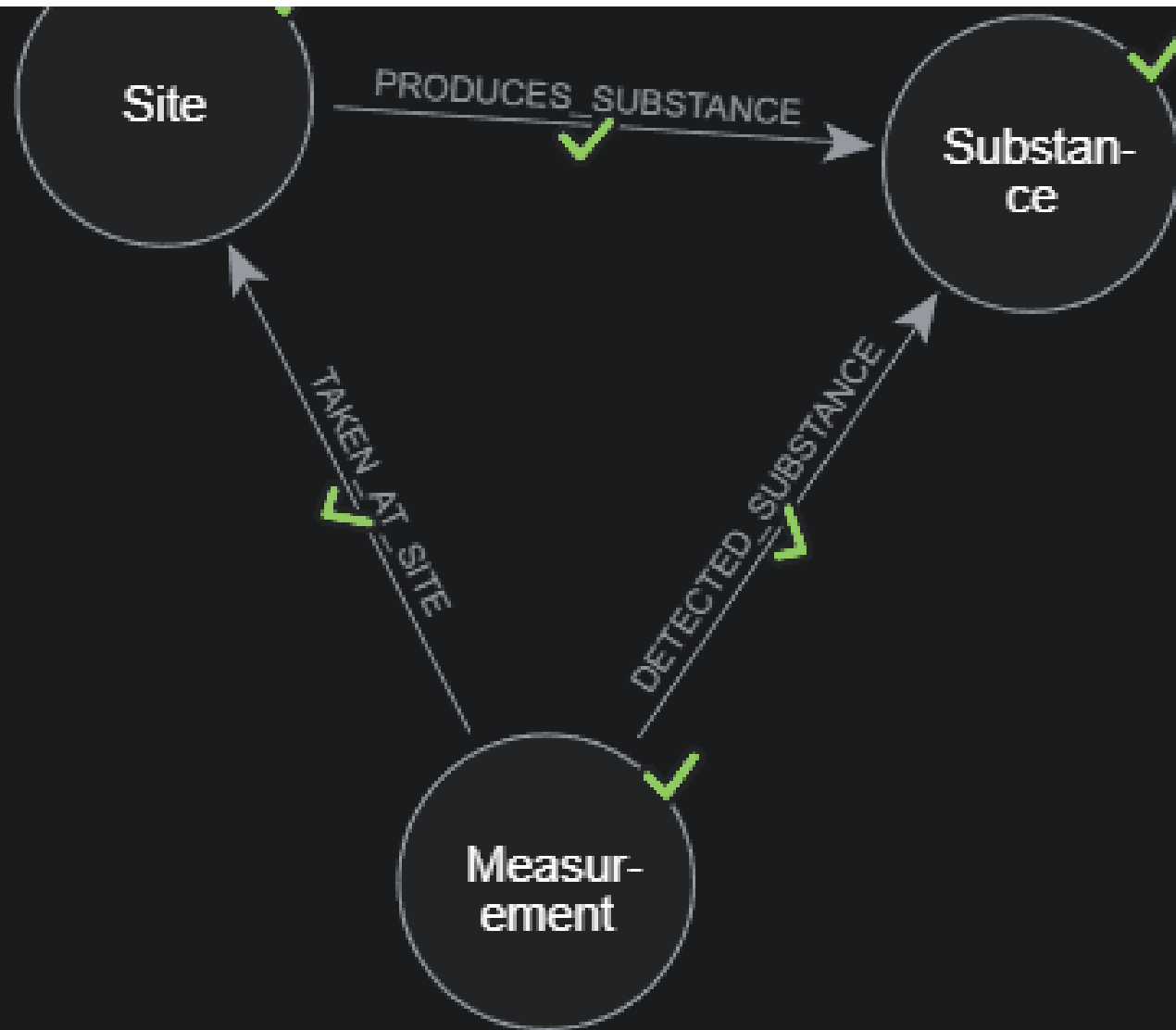
Fixed 127 chemical names, 9 map systems â†’ WGS84, all units â†’ ng/L, decoded hidden JSON.

â†’

Load

4 clean files: sites, samples, substances, connections â€” ready for Neo4j.





Schema screenshot from our project model: Site, Measurement, and Substance are linked with explicit graph relations.

Result: 87.3% completeness, 97.9% correct geolocation from a starting point of raw incompatible CSV files.

STEP 2 " PROPERTY GRAPH IN NEO4J

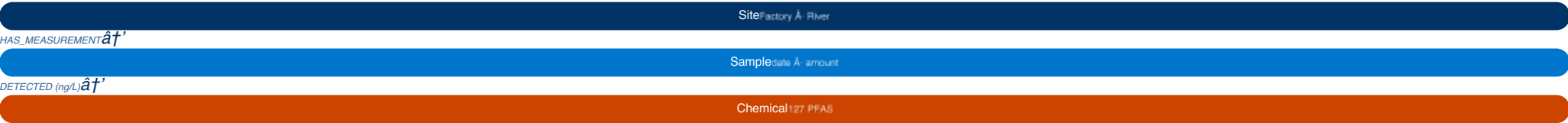
Spreadsheet limits

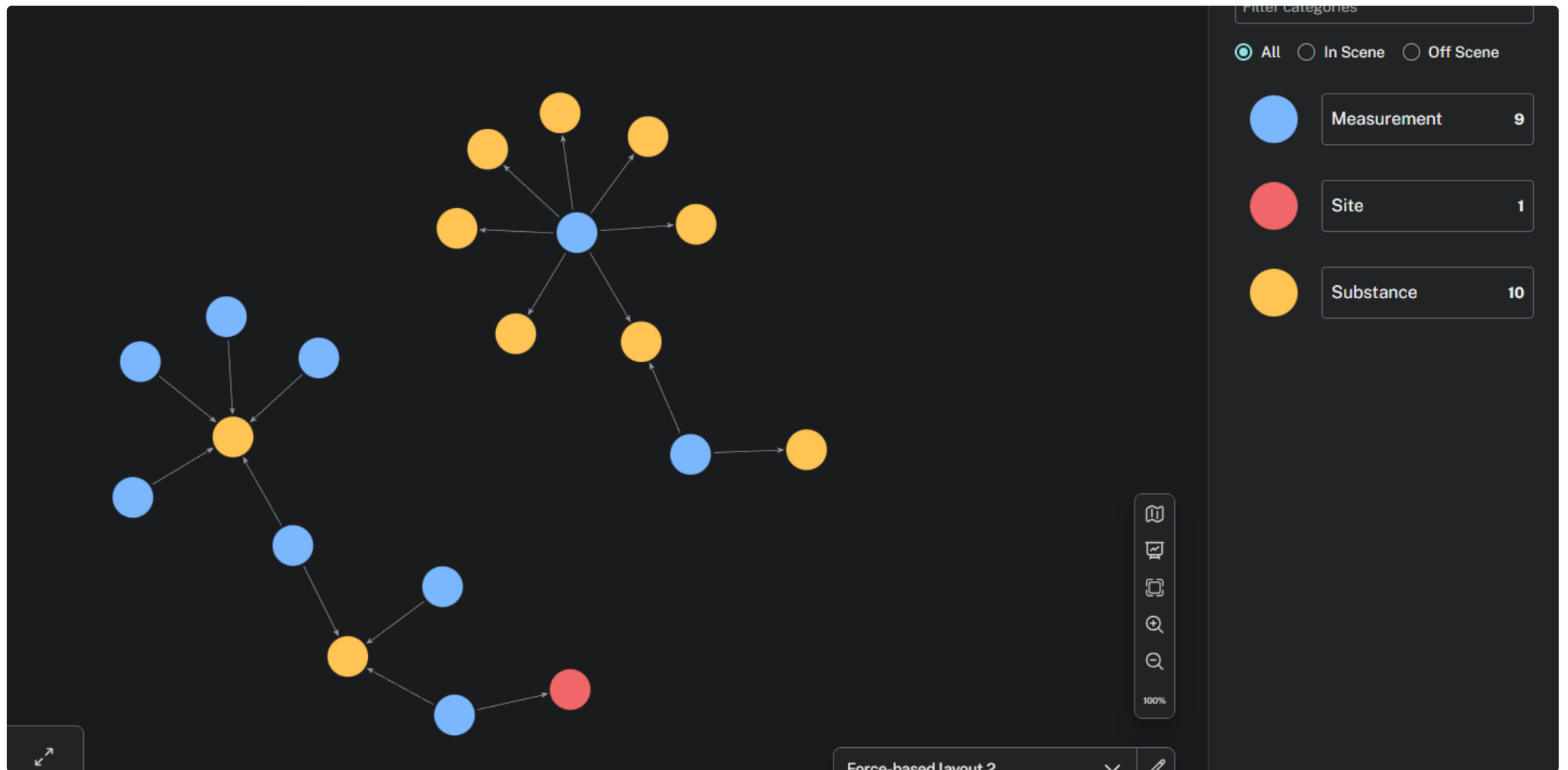
Cannot follow "factory - chemical - river - tap water" without 10+ JOIN operations and still misses hidden sites.

Property graph advantage

One query follows the full chain from any factory to any contaminated river, instantly.

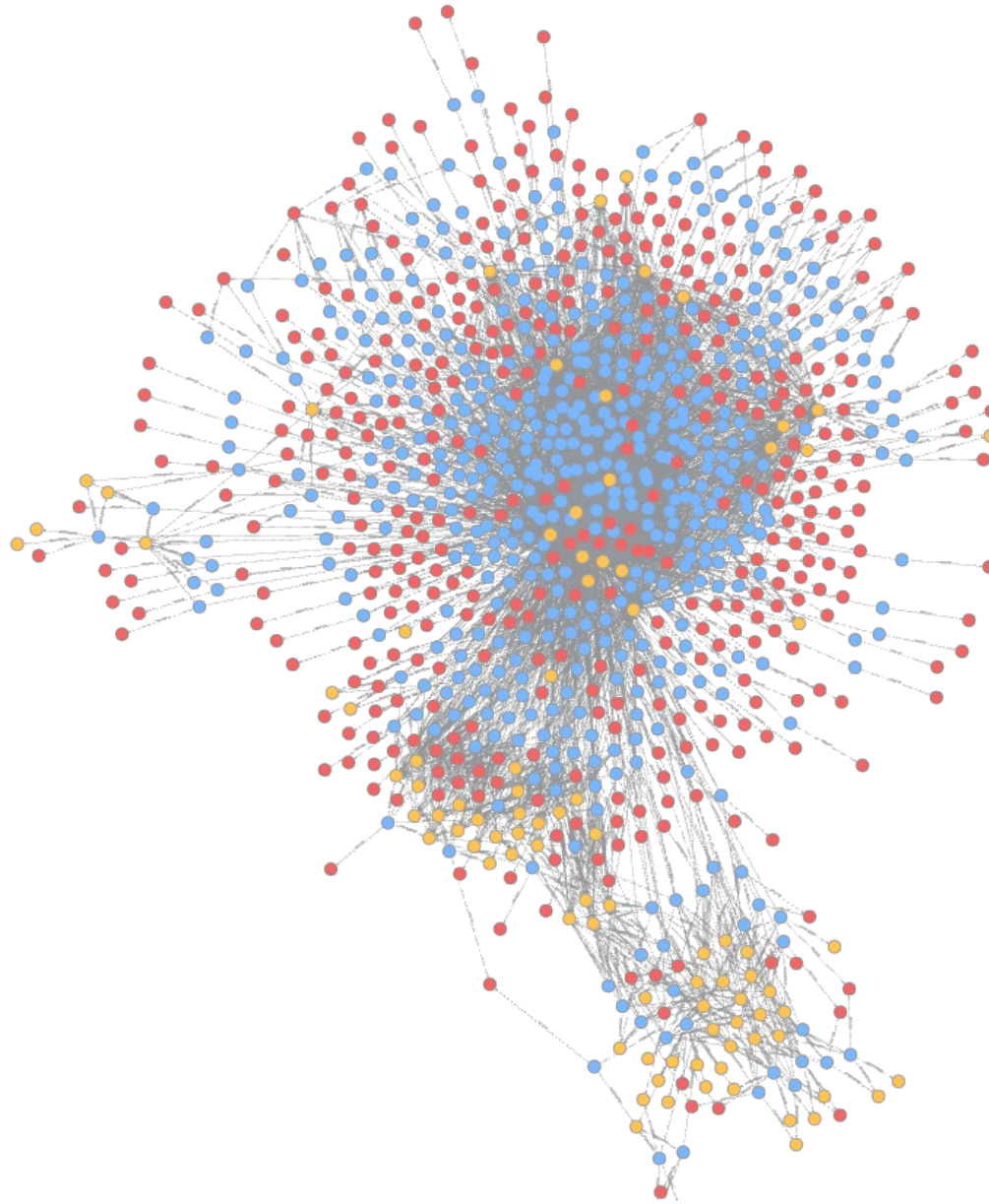
We used **Neo4j** (cloud graph database) with 3 connected node types:





Neo4j screenshot from projet-recherche: node colors identify the entities used in analysis.

Red: usine/site Blue: location/measurement point Yellow: substance found



Global system view (all data): Bloom visualization showing how the complete PFAS graph is explored at scale.

Lyon query â€” "Find most polluted sites"

```
MATCH (s:Site)-[:HAS_MEASUREMENT]->(m:Measurement)
WHERE s.city CONTAINS "Lyon" AND m.pfas_sum > 100
RETURN s.name, m.pfas_sum ORDER BY m.pfas_sum DESC
```

"Go to every Lyon site â†' follow to its measurements â†' return worst first."

STEP 3 - RISK SCORING (MCDA)

Q1

How many samples per site? â†' Impact Score I_k = log(sample count). Top: ZaCh System, 267 samples.

Q2

How dangerous is the site's name? â†' Textual Score T_k. "Arkema" in metadata â†' max score.

Q3

Reality check: follow graph to maximum concentration per site. Does score match real pollution?

Final evidence score per site:
Score = 0.4 - Reputation + 0.6 - Data volume
"Measures how well we know a site - not how polluted it actually is."

3 RESEARCH CONTRIBUTIONS - SUPERVISED BY PROF. BONIFATI (LIRIS, LYON 1)

Our work contributes to three active scientific domains. Prof. Angela Bonifati is one of Europe's leading experts in graph databases and co-author of the PG-Schema international standard.

Research 1 - Heterogeneous data integration

Problem: 104 datasets that look structurally identical but contain two incompatible ontologies hidden inside the same details column - called hidden semantic heterogeneity in database literature.

Contribution: applied bottom-up schema discovery - reverse-engineered what the data actually contained rather than imposing a fixed structure. Follows LIRIS research on DiscoPG (Bonifati et al., VLDB 2022).

Research 2 - Property graph modeling

Problem: SQL requires a fixed schema. Our data has 21.9% missing values that are not random (MNAR) and polymorphic JSON - both fatal for relational databases.

Contribution: designed a deterministic Property Graph schema (Sites - Measurements - Substances) natively handling optional properties, multiple node labels, and relation-level attributes (concentration on the DETECTED edge). Standard: PG-Schema (Angles, Bonifati et al., ACM 2023).

Research 3 - MCDA on graphs (novel)

Problem: standard MCDA (Saaty, 1980) assumes complete, homogeneous data. Ours has missing values and a mix of measured vs. suspected sites.

Contribution: adapted MCDA to run directly as a Cypher graph traversal - combining logarithmic data-density with qualitative textual scoring. MCDA has not previously been implemented as a native graph query. Validated on Lyon: found Salindres.

Why this is original research - in one sentence

Nobody had previously taken 104 incompatible European PFAS datasets, unified them through graph schema discovery, stored them in a Property Graph, and run MCDA directly on that graph to reveal hidden pollution hotspots. This is a first.

WHAT 17,500,000 NG/L ACTUALLY MEANS

Pollution scale - ng/L in water

01001,000100K17M+

Clean background water

<4 ng/L

EU legal limit (Jan 2026)

100 ng/L

Lyon tap water (Pierre-BÃ©nite)

up to 197 ng/L

Arkema â€” Pierre-BÃ©nite (rank 1st)

109,000 ng/L

GIE Chimie Salindres â€” hidden hotspot

17,500,000 ng/L

Salindres = 175,000Ã— the EU safe limit for drinking water.

MCDA RANKING â€” VALLÃ©E DE LA CHIMIE, LYON

Rank	Site & Location	Score	Real pollution
1st	Arkema France Location: Pierre-BÃ©nite, RhÃ¢ne (69) Seveso Ã— discharges into RhÃ¢ne	4.37	109,000 ng/L confirmed â€”
2nd	ZaCh System SA	4.17	267 samples PFAS confirmed 2023
Ranks 3â€”7: other Lyon-region sites â€” exact locations not in source data			
8th	GIE Chimie Salindres Location: Salindres, Gard (30340) Industrial chemical platform	3.80	17,500,000 ng/L Warning: 175,000Ã— EU limit

109,000 ng/L

Arkema France, Pierre-BÃ©nite â€” ranked 1st, confirmed â€”

Hidden hotspot â€” only found via the graph

17,500,000 ng/L

Salindres ranked 8th â€” low reputation, few samples. A spreadsheet sorted by score buries it. But the graph's query Q3 traced its actual measurements: 175,000Ã— the EU legal limit. Immediate intervention required.

The key contradiction

Our score measures *how well we know a site* â€” not how polluted it is. Salindres had low visibility but catastrophic reality. **The graph made this mismatch impossible to miss.**

WHAT THIS MEANS FOR PEOPLE AND NATURE

The Pierre-BÃ©nite area has had 3.5 tonnes of PFAS discharged into the RhÃ¢ne every year since 2011. That ends up in taps, soils, eggs, and fish. **Our graph makes it impossible to hide these connections.**

This tool can help scientists, governments and journalists prioritise which polluted sites need urgent clean-up across all of Europe.

- 1st European PFAS Property Graph
- MCDA in Cypher
- Bottom-up ETL audit
- Proof of concept â€” Auvergne-RhÃ¢ne-Alpes

Conclusion â€” what this means for people and nature

By turning 104 messy datasets into a clean knowledge graph, we built a tool that finds sites that **slip through every other detection method**. Pierre-BÃ©nite has had 3.5 tonnes of PFAS discharged into the RhÃ¢ne every year since 2011 â€” ending up in taps, soils, eggs, and fish. **Our graph makes these connections impossible to hide.**

How data engineering can help humanity

-
- Graph AI (GNN):** train AI on known contaminated sites to predict which untested areas are likely polluted â€” **before anyone gets sick**.
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- Real-time rivers:** connect graph to live sensors â€” contamination spikes detected in hours, not years.
-

Public health proof: link pollution data to hospital records to **prove causality** between a factory and cancer rates in a neighbourhood.

How data engineering helps nature

EU

European scale: apply the same pipeline to all 22,900+ contamination sites → one actionable map for environmental policy.

EC

Ecosystem protection: direct remediation resources to rivers and soils where they matter most → protecting wildlife and food chains.

LAW

Legal accountability: a connected graph gives regulators **traceable evidence** linking a factory to downstream contamination → usable in court.

```

# Full code

# rayanMELZI/
# Conversion-de-
# donnees-PFAS-du-CNRS
```